

A minimax free-energy account of prioritized reactivation and systems consolidation in coupled predictive-coding networks

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Abstract

During sleep and quiet wakefulness, the brain spontaneously revisits neural activity patterns associated with previous experience, a phenomenon known as reactivation. Reactivation is thought to support systems consolidation, the process by which memory retrieval becomes progressively more dependent on neocortical and less dependent on hippocampal circuits. Evidence indicates that reactivation is not indiscriminate but preferentially targets particular memory traces.

Systems consolidation has motivated an extensive modeling literature addressing spontaneous reactivation, replay prioritization, and the hippocampo-neocortical redistribution of memory representations. In parallel, a growing literature has described hippocampal and neocortical circuits within the predictive-coding (PC) framework. Here, we connect these two lines of work by showing how coupled PC networks can generate autonomous intrinsic activity that redistributes stored memory content between neural systems.

We model the hippocampo-neocortical system as two coupled PC associative memory networks. The asymmetric coupling is inspired by the interaction modalities observed between hippocampal area CA1 and the prefrontal cortex, combining bottom-up recruitment with effective top-down suppression. This coupling establishes a closed-loop dialogue in which hippocampal activity trains the neocortical network, while neocortical feedback biases subsequent hippocampal activity toward content that remains poorly represented.

Under simplifying assumptions and after neural activity has settled, the coupled dynamics identify the direction in the hippocampal memory manifold with the largest remaining neocortical reconstruction error. A subsequent local neocortical plasticity event changes the synaptic weights so as to produce the largest immediate reduction in this error. Repeated reactivation and plasticity therefore greedily minimize the largest representational gap between the two systems. As a direction becomes represented neocortically, the discrepancy that prioritized it declines, allowing reactivation to shift toward less well-represented content.

In simulations with correlated memories, the architecture transferred memory content faster than random memory rehearsal. Together, these results provide a PC account on how the neocortical influence could guide hippocampal reactivation, prioritizing content that remains weakly represented.

Author summary

1 Introduction

check for tense coherence

An important fraction of the brain’s metabolic budget is devoted to intrinsic activity, neuronal dynamics generated spontaneously by the brain in the absence of specific sensory input or behavioural demands [1, 2]. One of the best characterised events taking place during intrinsic activity is neural reactivation: the reinstantiation, during sleep or quiet rest, of activation patterns observed during waking experience (REF). For a substantial fraction of these events, activity in the hippocampus and neocortex is coordinated, giving rise to what is termed the hippocampo-cortical dialogue [3, 4].

This dialogue supports systems consolidation, the process by which memory retrieval becomes progressively less dependent on the hippocampus and more dependent on neocortical circuits [?, ?, 3, 5]. Systems consolidation is therefore often described as a form of memory transfer, whereby the neocortex progressively acquires memory content initially encoded in the hippocampus (REF).

Extensive literature indicates that the reactivation events underlying systems consolidation preferentially target certain memory traces over others. Traces that are recent, that correspond to novel experiences, or that **remain weakly encoded in cortex are prioritized for reactivation (REF)**. The precise neural mechanisms inducing this prioritization remain poorly understood (REF).

Reactivation and systems consolidation have motivated a broad range of computational models, from complementary learning systems [?] to autonomously reactivating attractor networks [?, ?, ?], generative replay [?, ?], and utility, context, or familiarity based replay scheduling [?, ?, ?, ?, ?]. These accounts differ in what drives reactivation, such as trace strength, recency, attractor competition, context, reward, or expected utility, and what information is transferred. Here, we focus on a single candidate mechanism, grounded in a specific communication motif between the hippocampus and neocortex.

The communication motif we consider has been most clearly characterized between the prefrontal cortex (PFC) and hippocampal area CA1, and combines two directional interactions: a bottom-up drive from CA1 to PFC and a top-down suppression from PFC to CA1. **During NREM sleep, a brief synchronized burst within a CA1 neuronal assembly is associated with increased firing in functionally coupled PFC populations (REF). Conversely, when PFC populations reactivate independently, without a coincident CA1 burst, CA1 neurons that normally participate in coordinated CA1–PFC events are preferentially suppressed (REF). Together, these findings motivate an asymmetric, representation-selective dialogue: hippocampal reactivation recruits selected prefrontal populations, whereas independent prefrontal reactivation suppresses the corresponding hippocampal representation. This motif may extend beyond PFC–CA1: entorhinal projections also recruit local inhibitory circuits in CA1, providing an potential additional pathway for similar bi-directional interaction (REF).**(change place maybe)

Wang, Poe, and Zochowski gave this motif a functional interpretation: cortical memory representations regulate hippocampal reactivation according to their degree of consolidation [6]. In their model, as a cortical representation forms during reactivation, it recruits hippocampal inhibitory interneurons that selectively suppress the corresponding hippocampal trace. As consolidation proceeds, each trace withdraws its own reactivation, leaving replay to the content that still lacks cortical support.

We extend this account in three ways. First, we develop the functional role of feedback inhibition in organizing memory redistribution. Second, we formulate the interacting hippocampal and neocortical memories within the predictive coding (PC) framework. Third, we show that, after both network states have converged, the resulting local plasticity rule takes a gradient step on a candidate objective for memory redistribution.

In the proposed model, neocortical feedback inhibition biases reactivation away from hippocampal content already supported cortically and toward content that remains poorly represented. This dynamic directs the consolidation process toward the largest representational gap between the two systems. We show that, within the PC framework and under simplifying assumptions, one neocortical plasticity event evaluated after activity convergence follows the negative gradient of the largest settled reconstruction error over the hippocampal memory manifold. The resulting architecture gives rise to prioritized reactivation, redistributing memory content as consolidation proceeds, and terminating reactivation once redistribution is complete.

More broadly, this work contributes to the study of fast and reliable offline information transfer

between artificial neural networks.

We deliberately adopt a simplified model in which the hippocampus and neocortex are each reduced to a single associative memory network of continuous activity units rather than spiking neurons; the resulting analytical tractability is what allows the mechanism to be derived rather than assumed. The analysis is performed on a linear version of the network, and the results are then generalized numerically for nonlinear activation functions.

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2 Models 60

In this section, we introduce the network architecture, beginning with its local dynamics and their coupling. We then present the network training and memory consolidation dynamics. 61
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The model consists of two coupled associative-memory PC networks (PCNs), the teacher and student networks, corresponding respectively to the hippocampus and neocortex. Taken independently, each network performs retrieval by carrying out inference on a given cue, minimizing its local free energy through changes in neuronal activity. Each network also learns by minimizing its local free energy, this time through changes in its synaptic weights (REF). The full architecture adds an asymmetric coupling between the teacher and student networks. 63
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Memory storage occurs in two successive phases. The teacher first acquires externally imposed activity patterns; the coupled network then transfers this content to the student during intrinsic consolidation (Sec. 2.3). 69
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2.1 A single predictive-coding associative memory 72

Tang et al. introduced the implicit covariance-learning PCN (implicit covPCN), which constitutes the basic building block of our architecture [7]. The implicit covPCN was developed as a more biologically plausible alternative to the explicit covPCN, whose learning rule requires non-local matrix operations and becomes numerically unstable on structured data [?, ?, ?, 7]. Here we present the linear version of the network which is the basis of the analytical study presented in this work. 73
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Both the teacher T and the student S , corresponding respectively to the hippocampus and neocortex, are implicit covPCNs with dense recurrent connectivity and no self-connections. Each performs associative memory through local inference and plasticity. 78
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We use $\alpha \in \{S, T\}$ to index either network. For activity $x_\alpha \in \mathbb{R}^d$ and recurrent weights W_α , define the prediction-error operator and error 81
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$$M_\alpha = I - W_\alpha, \quad \varepsilon_\alpha = M_\alpha x_\alpha, \quad (1)$$

and the free energy the network minimizes, 83

$$F_\alpha^{\text{rec}}(x_\alpha; W_\alpha) = \frac{\pi_\alpha}{2} \|\varepsilon_\alpha\|^2 = \frac{\pi_\alpha}{2} \|M_\alpha x_\alpha\|^2, \quad (2)$$

where $\pi_\alpha > 0$ is the precision of the prediction. We call F_α^{rec} the recurrent free energy because it accounts only for the network's internal prediction error generated by its recurrent connectivity. Network α lowers F_α^{rec} along two routes on two timescales: fast *inference* changes its activity, while slow *plasticity* changes its weights, 84
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$$\tau_\alpha \dot{x}_\alpha = -\pi_\alpha M_\alpha^\top \varepsilon_\alpha, \quad \dot{W}_\alpha = \eta \pi_\alpha \varepsilon_\alpha x_\alpha^\top, \quad (3)$$

with activity time constant τ_α and learning rate η . 88

The prediction may optionally pass through a pointwise nonlinearity f , giving $\varepsilon_\alpha = x_\alpha - W_\alpha f(x_\alpha)$. We take the linear case $f = \text{id}$ throughout the analysis and return to the rectified case $f = \text{ReLU}$ in Section ???. In addition, as introduced in the next section, the teacher activity is normalized such that $|x_T| = 1$ after each integration step. 89
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Each network is trained according to the plasticity dynamics in Eq. (3), either during interleaved clamping or through intrinsic activity during system consolidation (2.3). In a fully trained network α , the stored content is the kernel of its error operator, 93
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$$\mathcal{U}_\alpha = \ker M_\alpha = \{x \in \mathbb{R}^d : M_\alpha x = 0\}, \quad (4)$$

the network's memory manifold. Every stored pattern satisfies $M_\alpha x = 0$ and has zero recurrent free energy F_α^{rec} [7]. 96
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For the linear network, $\mathcal{U}_\alpha$ is a linear subspace. Any linear combination of stored memories is therefore a zero-energy fixed point. The network stores a flat memory manifold rather than a set of point attractors as in classical Hopfield networks (REF). 98
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2.2 Coupling the teacher and the student

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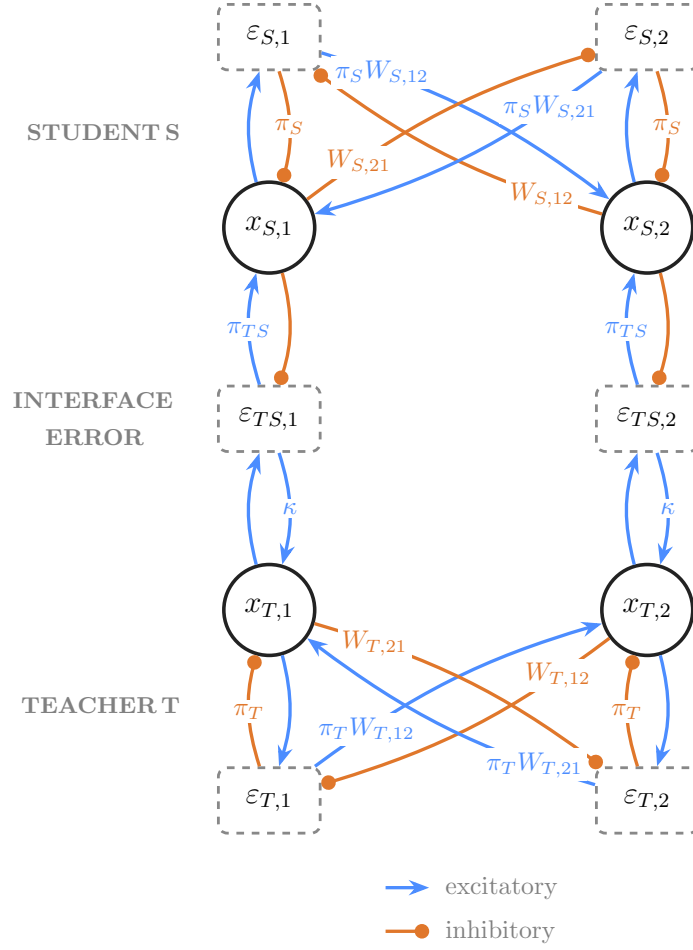


Fig 1. Model architecture (two units per population). Student S (top, cortical) above teacher T (bottom, hippocampal): value units x , recurrent errors $\varepsilon_S, \varepsilon_T$, and the one-to-one interface error $\varepsilon_{TS} = x_T - x_S$. Connections without labels have scalar value 1. During replay, the teacher-side interface gain κ makes the teacher ascend the student deficit rather than descending it as in classical PC networks.

Figure 1 shows the full architecture formed by coupling the student S and teacher T .

The two networks interact through a one-to-one *interface error*,

$$\varepsilon_{TS} = x_T - x_S, \quad (5)$$

which measures the mismatch between the student state and the teacher state currently expressed. Without the interface influence, both the teacher T and the student S activities follow the inference dynamics in Eq. (3) and descends their own recurrent free energy F_T^{rec} .

For the teacher, the coupling adds a direct interface-error drive, giving

$$\tau_T \dot{x}_T = \underbrace{-\pi_T M_T^\top \varepsilon_T}_{-\nabla_{x_T} F_T^{\text{rec}}} + \kappa \varepsilon_{TS}, \quad \|x_T\| = 1, \quad (6)$$

where κ is the teacher-side coupling gain. During intrinsic activity, $\kappa > 0$, so the interface-error population excites the teacher. In conventional predictive coding, minimizing the interface error with

respect to x_T would instead contribute $-\pi_{TS}\varepsilon_{TS}$ to the teacher dynamics. The teacher-side pathway therefore reverses the conventional sign. After each integration step, teacher activity is normalized to unit norm to prevent the linear dynamics from producing unbounded activity. This operation idealizes fast population gain control: in biological circuits, divisive normalization mediated by recurrent and feedforward inhibition constrains the overall response magnitude while largely preserving the relative activity pattern that carries the represented content [?].

The student side retains the conventional PC dynamics. Student activity minimizes its recurrent free energy and receives an excitatory correction from the interface-error population,

$$\tau_S \dot{x}_S = -\pi_S M_S^\top \varepsilon_S + \pi_{TS} \varepsilon_{TS}, \quad (7)$$

the student activity is not normalized during the simulation of the whole model.

Both contributions to Eq. (7) are gradient-descent terms on the student free energy functional which can therefore be expressed as

$$F_S(x_S, x_T; W_S) = \underbrace{\frac{\pi_S}{2} \|M_S x_S\|^2}_{F_S^{\text{rec}}} + \underbrace{\frac{\pi_{TS}}{2} \|x_T - x_S\|^2}_{F_{TS}}, \quad \tau_S \dot{x}_S = -\nabla_{x_S} F_S, \quad (8)$$

where $\pi_{TS} > 0$ is the precision of the interface prediction error. The student weights follow the local plasticity rule

$$\dot{W}_S = \eta \pi_S \varepsilon_S x_S^\top = -\eta \nabla_{W_S} F_S. \quad (9)$$

Activity and plasticity evolve concurrently in the simulations.

The overall influence between S and T deserve carefull reading. Although $\kappa > 0$ denotes an excitatory connection from the interface-error population to the teacher, the student enters $\varepsilon_{TS} = x_T - x_S$ with a negative sign, inhibiting the interface error. The combined pathway $x_S \rightarrow \varepsilon_{TS} \rightarrow x_T$ is thus inhibitory: increasing student activity reduces the excitatory error drive reaching the teacher. This effective top-down inhibition is the model counterpart of prefrontal suppression of hippocampal activity. In a conventional PC network, the interface error would instead inhibit the lower population., producing top down excitation through disinhibition.

[small paragraph with place for citation about why it is bio-plausible to normalize teacher activity]

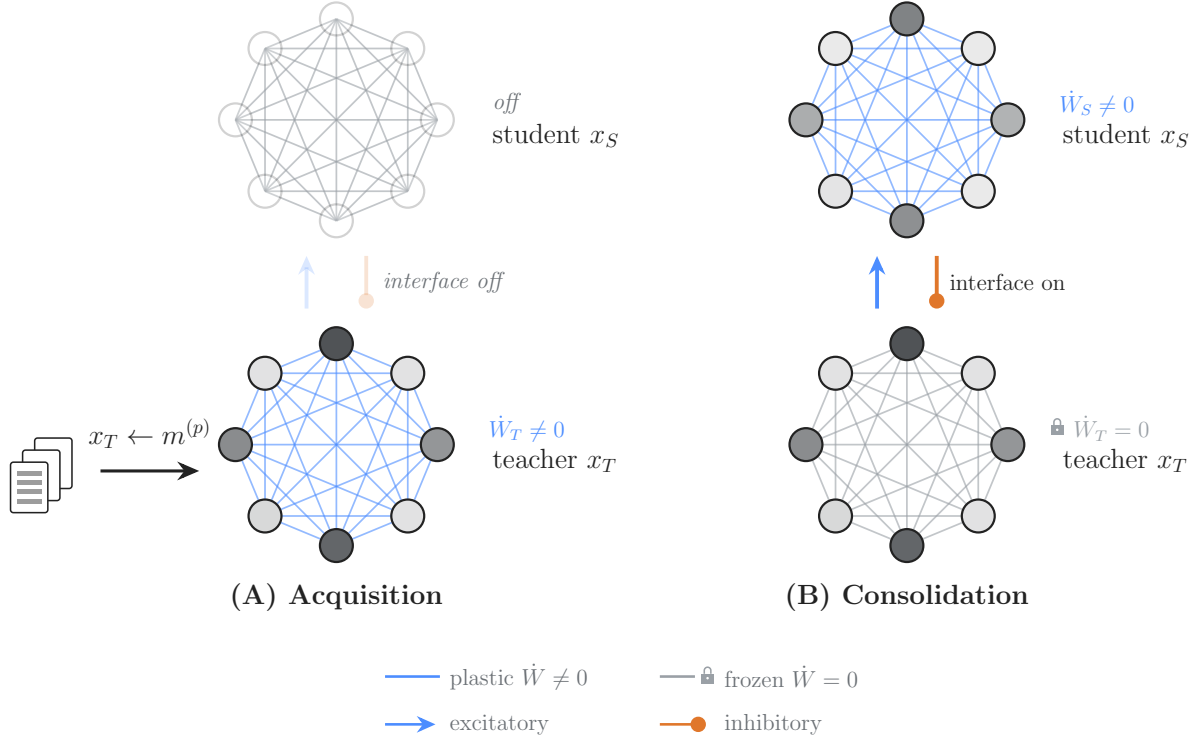


Fig 2. Initial storage and consolidation dynamics. Initial storage creates the teacher memory manifold from externally imposed patterns. Subsequent intrinsic dynamics use this manifold to generate student training states while preserving the teacher representation. The student is therefore trained by the coupled network dynamics rather than by direct clamping.

Memory is stored in the two networks by the same local recurrent plasticity rule (Eq. 3), but under two different regimes for generating neuronal activity. As shown in Fig. 2, the first phase establishes the teacher memory; the second redistribute (or transfer) this content to the student without presenting the student with an external target.

During initial storage (Fig. 2(A)), teacher activity x_T is clamped to memory patterns $m^{(p)}$ in interleaved order while W_T follows Eq. 3. The student and the interface remain inactive. These updates reduce the teacher's recurrent prediction error on the presented patterns and establish its memory manifold $\mathcal{U}_T = \ker M_T$ (Methods). Interleaved training allows correlated memories to be stored while mitigating interference (REF). In contrast to our model, the hippocampus does not appear to require interleaved training to reliably store memory content during awake activity (REF). The orthogonalization of memory representations, particularly in the granule-cell layer of the dentate gyrus, is thought to support rapid learning of correlated memory traces (REF). This biological machinery is not part of our model and is therefore replaced by interleaved training.

During consolidation (Fig. 2(B)), the external clamp is removed and the interface is activated. The teacher and student activities follow Eqs. (6) and (7), while W_T remains fixed and only W_S follows the plasticity rule in Eq. (9). The student is never clamped: teacher-driven intrinsic activity supplies the states stored by its local recurrent plasticity and influence the student through the interface error.

3 Results

3.1 Reactivation and memory content redistribution

We first examined whether intrinsic activity in our model could reactivate and redistribute stored memory content.

We stored three unit-normalized MNIST patterns in a linear 784-unit teacher network, froze the teacher’s recurrent weights, and started the coupled dynamics with a plastic student whose recurrent weights were initialized to zero. During this intrinsic phase, neither network received external input. Low-amplitude Gaussian noise was applied to the teacher activity, allowing the dynamics to explore the teacher memory manifold. The results are shown in Fig. 3.

Fig. 3(A, bottom), (B, top), and (C) show that teacher activity is dominated by mixture states rather than by reactivation of individual patterns. This behavior follows from the linear activation function: the stored patterns span a flat memory manifold containing every signed linear combination of those patterns, including their antipodal states.

Fig. 3(D) and (E) provide direct evidence of memory-content redistribution during intrinsic activity. The successive reduction of the three eigenvalues indicates that the student acquires the flat teacher memory manifold stages by stages across independent directions, rather than through a uniform reduction of error across all directions. redistribution rate slows once most of the manifold is represented by the student and remains incomplete over the displayed interval. Longer simulations further reduce the residual redistribution gap, although at a much lower rate. At this late stage, the mismatch-driven selection signal is weak and the teacher increasingly relies on Gaussian-noise exploration to sample the remaining poorly represented directions. Despite this incomplete redistribution, the isolated-student probes in Fig. 4 show that, after our consolidation protocol, the student successfully store the original patterns. Memory reconstruction capacities of the student are uneven at the partial checkpoint but allow recognizable reconstruction of all three memories later during consolidation.

Fig. 3(B, bottom) and (E, middle and bottom) show how the signals driving learning and reactivation attenuate during memory redistribution. The student plasticity rate decreases as the student–teacher mismatch contracts, revealing a self-limiting learning process: as the student acquires teacher-supported content, it removes the errors that generate further weight updates. In parallel, the teacher-side selection drive, generated by the mismatch-dependent top-down influence of the student on the teacher, also decreases. Three transient peaks accompany the successive reduction of the three error modes in Fig. 3(D), consistent with the sequential acquisition of the three independent directions spanning the rank-three teacher memory manifold. Both signals remain nonzero at the end of the displayed interval, so this run approaches but does not reach exact self-extinction.

To test whether this mechanism extends beyond redistribution from a fixed teacher memory set, we allowed the teacher repertoire to grow during an otherwise continuous consolidation run (Fig. 5). Each newly added memory introduced a new high-eigenvalue error mode, while the eigenvalues associated with memory directions already redistributed to the student remained close to zero. These new gaps then declined without resetting the student or discarding its previously learned weights, showing that the same mismatch-dependent dynamics can repeatedly redirect reactivation toward newly unsupported content. This protocol captures a feature of systems consolidation: ongoing experience continually adds new hippocampal memory content, which must then be integrated into neocortical circuits during subsequent periods of intrinsic activity (REF). In the model, each newly stored hippocampal direction automatically recreates a hippocampo-cortical mismatch. Renewed prioritization therefore requires neither an explicit replay queue nor an externally assigned novelty tag. As cortical support for the new content develops, both the mismatch and the associated learning drive recede.

In summary, the coupled architecture redistributes memory content from the teacher to the student during intrinsic activity. redistribution remains incomplete in the displayed run, yet the isolated-student probes show recognizable completion of all three stored patterns at the late checkpoint. Exact equality between the teacher and student memory manifolds is therefore not required for successful recall in this simple task. This result should nevertheless be interpreted as evidence of functional recall in a small associative-memory setup, rather than as a general demonstration of functional equivalence between the student and the teacher networks for more complex tasks. Tests involving classification of the

reconstructed patterns, weaker cues, or more complex memory sets will be needed to determine when the residual redistribution deficit becomes functionally limiting. In the Discussion, we consider possible mechanisms for improving redistribution exhaustiveness.

The observed dynamics are consistent with neocortex-dependent prioritization. The mismatch signal biases teacher activity toward teacher-supported directions that remain poorly represented by the student, and this bias weakens as those directions are learned. This provides a possible mechanistic account of the preferential reactivation of recently encoded hippocampal representations during post-learning sleep (REF), followed by reduced hippocampal involvement as memory retrieval becomes increasingly supported by neocortical circuits (REF). Nevertheless, the model does not treat recency itself as the priority signal. It predicts that reactivation depends more directly on the mismatch between hippocampal support and cortical representation.

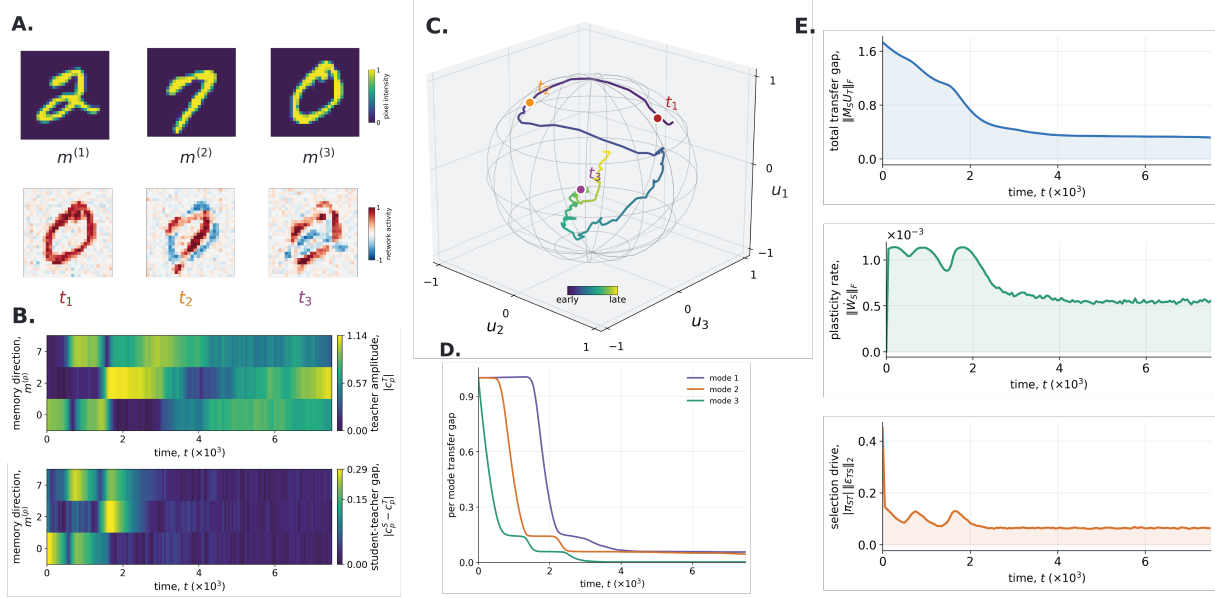


Fig 3. Spontaneous intrinsic activity progressively redistributes teacher-supported memory content. The frozen teacher network was coupled to a student network initialized with zero activity and zero recurrent weights. No external input was presented during the run; low-amplitude Gaussian noise seeded exploration of the teacher memory manifold. **A**, The three stored MNIST patterns (top; display order 2, 7, and 0) and teacher states at $t_1 = 250$, $t_2 = 1000$, and $t_3 = 7000$ (bottom). Although the stored patterns are positively signed, the linear network can express signed mixtures and antipodal states. **B**, (Top) Memory coefficient magnitude $|c_p^T(t)|$, which measure the contribution of each stored pattern $m^{(p)}$ to teacher activity $x_T(t)$ (Method). Low values mean the teacher doesn't reactivates the memory while large values indicate strong reactivation. (Bottom) Student-teacher magnitude gaps $|c_p^S(t) - c_p^T(t)|$; a large gap indicates that the two networks express different magnitude along that pattern coordinate. **C**, Teacher activity projected onto an orthonormal basis U_T of the three-dimensional teacher memory space. The wireframe marks the unit sphere imposed by teacher normalization, and the colored markers identify the snapshots in panel A. Basis axes are orthogonal memory-space directions, not individual digits. **D**, Sorted eigenvalues of $G_S(t) = U_T^T M_S(t)^T M_S(t) U_T$. High eigenvalues identify teacher-supported directions poorly represented by the student, whereas low values indicate accurate storage of these components. **E**, (Top) Total redistribution gap $\|M_S U_T\|_F$, which measures how much of the teacher memory space remains unsupported by the student. (Middle) Student plasticity rate $\|\dot{W}_S\|_F$, the magnitude of the current recurrent-weight update. (Bottom) Teacher-side selection drive $|\pi_{ST}| \|\varepsilon_{TS}\|_2$, the magnitude of the mismatch-dependent signal acting on the teacher.

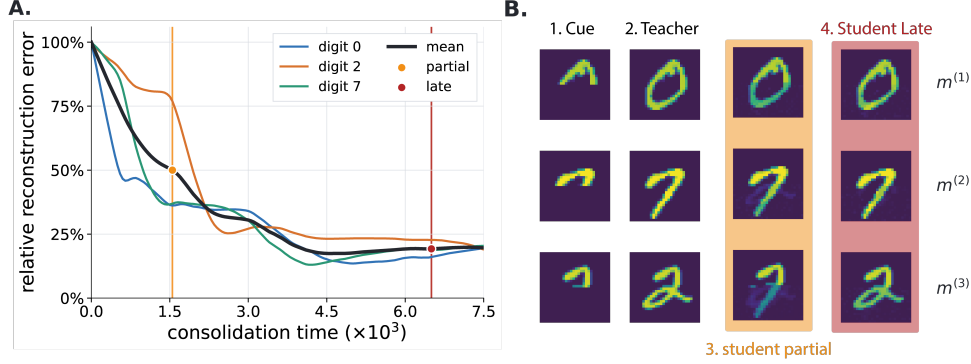


Fig 4. Transferred teacher memories become reconstructable by the isolated student. **A**, Relative recurrent reconstruction error for each stored MNIST pattern during consolidation. At each recorded time, the student weights were frozen, its state was set to the complete pattern $m^{(p)}$, and the error was evaluated as $r_p(t) = \|m^{(p)} - W_S(t)m^{(p)}\|_2 / \|m^{(p)}\|_2$. Thus, 100% corresponds to the initially empty student with $W_S = 0$, whereas zero denotes exact recurrent prediction of that pattern. The black curve is the mean over the three digits; orange and red markers identify the partial ($t = 1550$) and late ($t = 6500$) checkpoints used in panel B. **B**, Clamped recall by isolated, frozen networks. From left to right, each row shows the partial cue, the teacher reconstruction, and student reconstructions at the partial and late checkpoints. The upper half of each digit was clamped to the stored pixels, the hidden half was initialized to zero, and the free units settled by minimizing the network self-energy $\frac{\pi}{2} \|(I - W_\alpha)x_\alpha\|_2^2$, with no teacher-student coupling or ongoing plasticity. Student recall performances are uneven at the partial checkpoint, in agreement with the reconstruction error in panel A. For the late checkpoint, all reconstruction outputs visually approach the teacher completions despite a remaining relative reconstruction error.

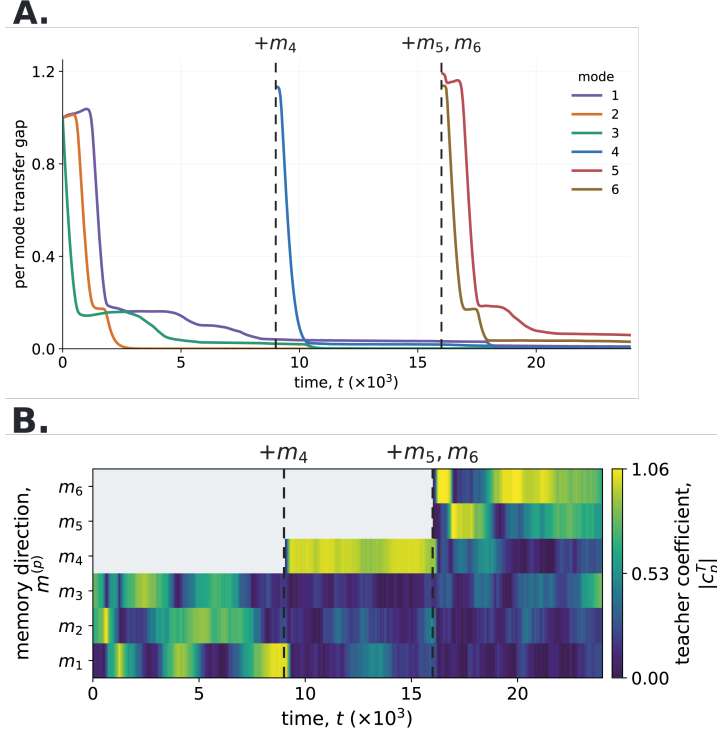


Fig 5. Staged memory acquisition renews reactivation and memory redistribution. A 50-unit linear teacher-student network was followed through one continuous run with six unit-normalized random patterns. The teacher initially stored m_1 , m_2 , and m_3 ; it acquired m_4 at $t = 9000$ and acquired m_5 and m_6 together at $t = 16000$ through interleaved teacher training on the whole corpus. Coupled intrinsic dynamics resumed after each acquisition without resetting student activity or recurrent weights. Dashed lines mark the acquisition events. **A**, Eigenvalues of the student error operator restricted to the currently supported teacher memory space. Previously transferred modes keep a small error, whereas each newly supported direction introduces a large transfer gap that subsequently decreases. **B**, Memory coefficient magnitudes $|c_p^T(t)|$, which measure the contribution of each stored pattern $m^{(p)}$ to teacher activity $x_T(t)$ (Methods). Low values mean that the teacher does not reactivate the corresponding memory, whereas high values indicate strong reactivation. Newly added memories initially receive a stronger reactivation drive. As redistribution progresses, this drive decreases.

3.2 Fast memory content transfer of structured data

Having established that the coupled network dynamics can redistribute memory spaces, we next asked how efficiently this redistribution proceeds when the corpus contains unequally sized groups of correlated memories. We compared two protocols:

- **Intrinsic transfer (our method).** A trained, frozen teacher storing the memory space was coupled to an initially empty student. Teacher and student activity evolved under the coupled intrinsic dynamics, and the student updated its recurrent weights after each integration step.
- **Stochastic rehearsal.** At each plasticity event (time step), one memory was sampled uniformly from the indexed corpus and clamped directly onto the student before the same local recurrent update was applied.

Stochastic rehearsal emulates the redistribution dynamics occurring from non-prioritized reactivation of memories, comparing it with our method therefore allows us to assess the impact of prioritized reactivation on the redistribution dynamics.

The results for a correlated memory corpus are shown in Fig. 6.

We considered a dataset of 26 memories divided in three mutually orthogonal groups containing 18, 5, and 3 memories. Fig. 6(A) display the correlation structure of the memory set, memories within a group had a correlation of $\rho = 0.70$. The corpus construction, matched parameters, and implementation details are given in Materials and methods, Section 6.6.

We followed redistribution at two levels. The named-memory residual used in the heatmaps asks whether the student can reconstruct each stored example. We also reported a memory-subspace reconstruction error, R_{sub} , which give direct insight on how well the student network actually store the wanted memory manifold. We also checked, the worst-direction reconstruction error, R_{worst} , which isolates the single direction that the student store least accurately. Formal definitions of all three measures are given in Materials and methods, Section 6.6.

The two protocols initially reduced the emory-subspace reconstruction error at similar rates, but intrinsic transfer soon produced a faster decline than stochastic rehearsal (Fig. 6(B)).

Fig. 6(C) shows that intrinsic transfer reduced reconstruction errors at similar rates across all three groups, despite their unequal sizes, whereby stochastic rehearsal favor learning for the large clusters.

The difference in transfer rates reflects how each protocol selects the activity that drives plasticity. Stochastic rehearsal samples named memories uniformly. Because memories within each group are strongly correlated, these updates repeatedly reinforce their shared component. This induces faster learning of over-represented groups. On the other hand, intrinsic transfer is not tied to named-memory frequency. As the student learns a teacher-supported direction, the mismatch along that direction contracts and its selection drive weakens, allowing activity to shift toward directions with larger remaining deficits. The result is consistent with deficit-dependent reactivation allocating plasticity according to reconstruction error of the subspace rather than the sample size of a given correlated memory-cluster.

Fig. 7 shows how within-group correlation affects the redistribution rate of both protocols while memory group sizes remain fixed. At low correlation, stochastic rehearsal was faster, whereas increasing correlation shifted the advantage toward intrinsic transfer.

Stochastic rehearsal can be more efficient when correlations are weak. Because the corpus acts as a lookup table, each plasticity event is applied directly to an exact memory. Intrinsic transfer instead updates the student while activity moves through the teacher memory space, and these intermediate states may produce partly redundant updates. Direct presentation therefore retains an advantage when uniform sampling already covers the memory subspace evenly, which happens when correlation fall.

Together, these results show that deficit-dependent intrinsic dynamics can redistribute highly correlated memory content with fewer plasticity events than stochastic rehearsal. This advantage grows with correlation, when uniform sampling increasingly produces overlapping updates. Such efficiency would matter for the brain as energy and opportunities for offline reactivation are limited (REF).

The next section presents a mathematical account of this qualitative behavior. We ask whether the coupled dynamics can be understood as optimizing an objective function derived from the PC formulation.

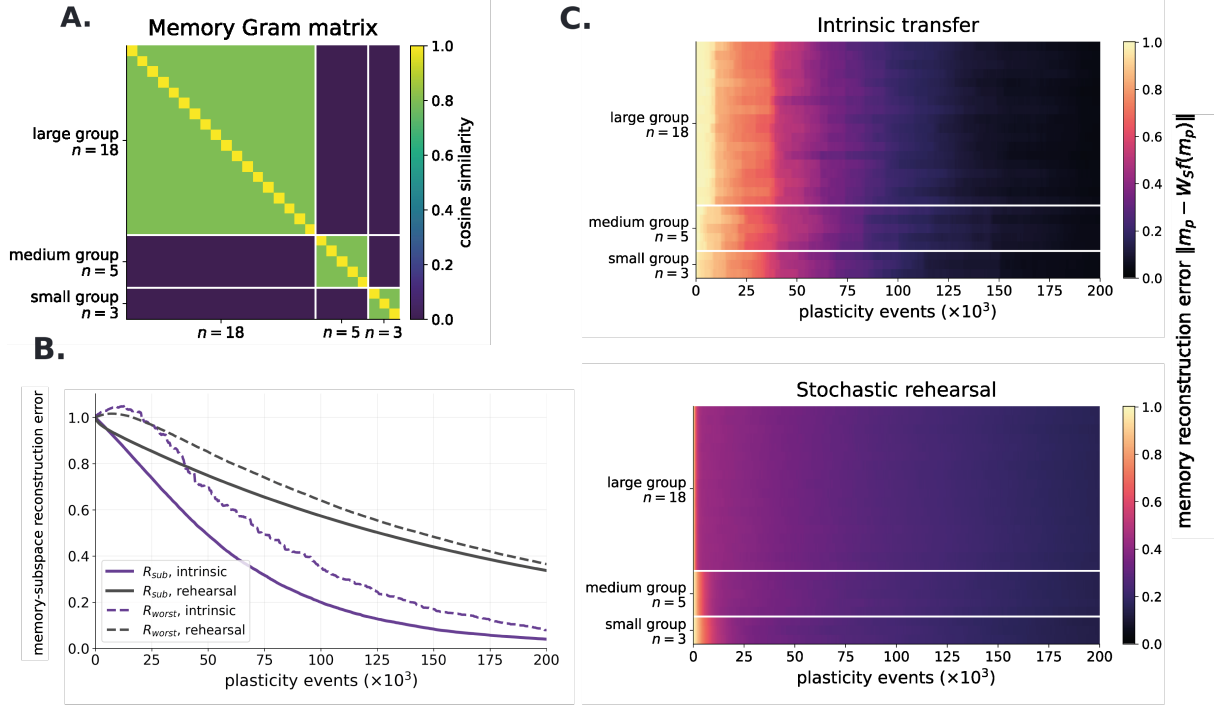


Fig 6. Intrinsic transfer accelerates redistribution across unequal groups of correlated memories. Both conditions used linear students initialized with $W_S = 0$, the same memory corpus, the same recurrent plasticity rule and rate, and 200,000 plasticity events. The corpus contained 26 memories divided into three mutually orthogonal groups of 18, 5, and 3 memories, with exact within-group cosine similarity $\rho = 0.80$. **A**, Memory Gram matrix. Each cell shows the cosine similarity between two memories. **B**, Reconstruction error over the complete memory space. Solid curves show the RMS memory-subspace error R_{sub} , which gives equal weight to every independent direction. Dashed curves show the worst-direction error R_{worst} , defined by the memory-space direction reconstructed least accurately by the student. Both measures equal one for the initially empty student, and lower values indicate more complete redistribution. After a short initial transient, intrinsic transfer reduced both errors faster than stochastic rehearsal and reached substantially lower final values. The rehearsal errors were still decreasing at the end of the protocol. **C**, Reconstruction errors for individual memories during intrinsic transfer (top) and stochastic rehearsal (bottom). Light colors indicate high reconstruction error and dark colors indicate accurate reconstruction. Intrinsic transfer reduced errors at similar rates across all three groups, whereas under stochastic rehearsal, errors remained higher for longer in the smaller groups. Metric definitions and full protocol details are provided in Materials and methods, Section 6.6.

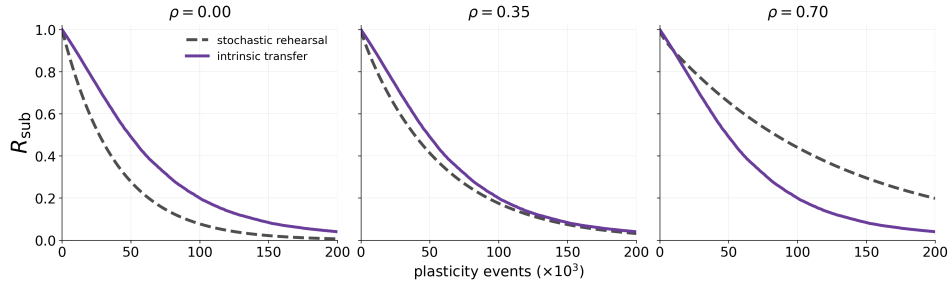


Fig 7. Increasing within-group correlation shifts the advantage from stochastic rehearsal to intrinsic transfer. The three conditions used the same group sizes (18, 5, and 3), student initialization, plasticity parameters, and number of plasticity events. Only the exact within-group cosine similarity changed, with $\rho \in \{0, 0.35, 0.70\}$. Each curve shows the RMS memory-subspace reconstruction error R_{sub} , with lower values indicating more complete redistribution. Gray dashed curves show stochastic rehearsal and purple curves show intrinsic transfer. At $\rho = 0$, stochastic rehearsal reduced the error faster. At $\rho = 0.35$, the two protocols performed similarly. At $\rho = 0.70$, intrinsic transfer became faster after a brief initial period and reached a substantially lower final error. As correlation increased, stochastic rehearsal slowed because uniformly sampled memories produced increasingly overlapping updates, whereas the intrinsic-transfer curves changed comparatively little. All conditions used linear units. Full definitions and protocol details are provided in Materials and methods, Section 6.6.

3.3 Approximating minimization of the largest settled neocortical free energy

Draft status: the analytical result below is derived from the linear model. The numerical tests and figure are proposed protocols; their outcomes have not yet been measured.

The preceding results show that the coupled dynamics can redistribute memory content from the teacher to the student. We now ask whether redistribution progressively reduces the largest settled student free energy over the teacher memory manifold. We first show that, after student relaxation, the teacher-side interface drive is proportional to the gradient of the settled student free energy with respect to teacher activity. This result motivates two hypotheses:

1. **Activity selection:** the coupled dynamics approach a teacher state that nearly maximizes the settled student free energy over the teacher memory manifold.
2. **Synaptic plasticity:** a local student-plasticity event applied after activity relaxation approximates a gradient-descent step on this maximum.

Figure 8 outlines the proposed numerical tests of these hypotheses.

The following analysis and proposed numerical tests consider the separation of timescales

$$\tau_S \ll \tau_T \ll \frac{1}{\eta}. \quad (10)$$

The first inequality allows student activity to relax while teacher activity changes little. This permits adiabatic elimination of the student activity, replacing it by its equilibrium for the current teacher state. The second inequality keeps synaptic weights approximately fixed during activity relaxation. The analytical derivation presented here uses the adiabatic limit, while the numerical tests assess this approximation at finite timescale separation.

As defined in Eq. (8), the student free energy F_S combines the student's recurrent prediction error with the mismatch between the teacher and student states. For fixed teacher activity x_T and student weights W_S , the linear student dynamics minimize F_S with respect to x_S . We denote the free energy remaining after this relaxation by

$$q(x_T; W_S) = \min_{x_S} F_S(x_S, x_T; W_S). \quad (11)$$

Under the adiabatic approximation, q therefore describes how the settled student free energy varies as teacher activity changes.

The relevant set of teacher states is the unit sphere of its memory manifold,

$$\mathbb{S}_T = \{x_T \in \mathcal{U}_T : \|x_T\| = 1\}, \quad \mathcal{U}_T = \ker M_T. \quad (12)$$

The two parts of this constraint are implemented differently in the network. Renormalization after each integration step enforces unit norm. **Membership in $\mathcal{U}_T$ is approximate: the teacher's recurrent dynamics damp activity outside its memory manifold, but the interface drive and noise can produce departures from it. We therefore use $\mathbb{S}_T$ to define the optimization benchmark and measure the actual teacher's distance from this set.**

to change when we have the data

The largest settled student free energy over $\mathbb{S}_T$ is

$$\mathcal{F}_{\max}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} q(x_T; W_S) = \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_S(x_S, x_T; W_S). \quad (13)$$

By definition,

$$q(x_T; W_S) \leq \mathcal{F}_{\max}(W_S; W_T) \quad \text{for every } x_T \in \mathbb{S}_T.$$

Thus, $\mathcal{F}_{\max}$ bounds the settled student free energy over the entire teacher memory manifold. A low value means that the student settles to a low free energy for every state on this manifold, whereas a large value means that at least one state remains poorly represented.

Hypothesis 1 therefore predicts that, during intrinsic activity with fixed student weights W_S , the teacher approaches a state whose settled student free energy is close to this upper bound. We test this prediction by comparing the observed teacher state with an independently calculated ideal teacher state that attains the maximum.

This nested quantity also has an interpretation in terms of the complete teacher–student network free energy,

$$F_{\text{net}}(x_T, x_S; W_T, W_S) = F_T^{\text{rec}}(x_T; W_T) + F_S(x_S, x_T; W_S). \quad (14)$$

Since $\mathcal{U}_T = \ker M_T$, the teacher recurrent free energy vanishes for every $x_T \in \mathbb{S}_T$. Consequently,

$$\mathcal{F}_{\text{max}}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_{\text{net}}(x_T, x_S; W_T, W_S) \quad (15)$$

The largest settled student free energy is therefore also the largest settled free energy of the complete network over this manifold.

We call a *redistribution cycle* the relaxation of teacher and student activity at fixed weights followed by one student-plasticity event. Hypothesis 2 predicts that, once the teacher has approached a state maximizing the settled student free energy and the student has settled in response to that state, the local update of the student weights approximates a gradient-descent step on $\mathcal{F}_{\text{max}}$:

$$\boxed{\Delta W_S \approx -h P_0 \nabla_{W_S} \mathcal{F}_{\text{max}}(W_S; W_T)}, \quad h = \eta \Delta t_{\text{pl}}, \quad (16)$$

where Δt_{pl} is the event duration and P_0 sets the diagonal entries of a matrix to zero, corresponding to the absence of self-connections in our model.

This construction is close in purpose to classical predictive coding, while differing in how the states on which free energy is minimized are obtained. PC networks ordinarily minimize free energy for activity elicited by current sensory input; variation in those inputs supplies the states on which learning acts. During autonomous redistribution, no external input specifies the state to be learned. The network must therefore organize both exploration of the teacher memory manifold and local student learning. Exploration may temporarily increase the instantaneous network free energy, while successive student-plasticity events are expected to reduce the network settled free energy over the teacher memory manifold.

The settled interface error is proportional to the gradient with respect to teacher activity. The following result identifies how the student’s equilibrium response determines the interface drive received by the teacher. Define

$$S_S = M_S^\top M_S, \quad H_S = \pi_{TS}I + \pi_S S_S, \quad N_S = I - \pi_{TS}H_S^{-1} = \pi_S S_S H_S^{-1}. \quad (17)$$

Theorem 1 (Settled student response and interface-driven ascent). *For the linear student with $\pi_S, \pi_{TS} > 0$, fixed teacher activity, and fixed weights, F_S has the unique global minimizer*

$$x_S^*(x_T) = \pi_{TS}H_S^{-1}x_T = (I - N_S)x_T. \quad (18)$$

The settled student free energy and its gradient with respect to teacher activity are

$$q(x_T; W_S) = \frac{\pi_{TS}}{2}x_T^\top N_S x_T, \quad \nabla_{x_T} q = \pi_{TS}N_S x_T = \pi_{TS}(x_T - x_S^*). \quad (19)$$

Thus, when the student is at equilibrium and $\kappa > 0$, the teacher-side interface contribution is a positive multiple of this gradient:

$$\kappa \varepsilon_{TS}^* = \frac{\kappa}{\pi_{TS}} \nabla_{x_T} q. \quad (20)$$

Proof. For fixed x_T , the student gradient and Hessian are

$$\nabla_{x_S} F_S = H_S x_S - \pi_{TS} x_T, \quad \nabla_{x_S}^2 F_S = H_S \succ 0.$$

The positive-definite Hessian ensures that student gradient flow at fixed teacher activity converges to $x_S^*(x_T)$. Solving the stationarity equation gives Eq. (18). Expanding the energy and using $H_S x_S^* = \pi_{TS} x_T$ gives

$$q = \frac{\pi_{TS}}{2}x_T^\top x_T - \frac{\pi_{TS}}{2}x_T^\top x_S^* = \frac{\pi_{TS}}{2}x_T^\top N_S x_T.$$

Since N_S is symmetric, differentiating it gives $\nabla_{x_T} q = \pi_{TS}N_S x_T$. Finally, $x_T - x_S^* = N_S x_T$, which identifies the gradient with the settled interface error. $\square$

The theorem shows that, once the student has settled, the teacher-side coupling favors changes in teacher activity that increase the settled student free energy. Within the teacher memory manifold, this favors states that remain poorly represented by the student. This result alone does not establish that the full coupled dynamics approach a teacher state whose settled student free energy is close to the maximum over $\mathbb{S}_T$. This is the prediction of Hypothesis 1, which we test numerically below.

Testing Hypothesis 1: selection of teacher states with near-maximal settled student free energy. For fixed teacher and student weights, an *ideal teacher state* x_T^* is a unit state on the teacher memory manifold for which the student’s equilibrium response leaves the largest free energy:

$$x_T^* \in \arg \max_{x_T \in \mathbb{S}_T} q(x_T; W_S), \quad q(x_T^*; W_S) = \mathcal{F}_{\max}(W_S; W_T). \quad (21)$$

This is a reference for teacher activity; the corresponding student state is its response $x_S^*(x_T^*)$, which generally differs from x_T^* . We calculate the ideal teacher state directly from the fixed weights, independently of the simulated trajectory. The calculation is given in Materials and methods, Section 6.5.1.

In the linear model, x_T^* and $-x_T^*$ yield the same settled student free energy. When these are the only maximizing states, we measure the distance from the observed teacher state to the closer of the two:

$$d_T = \min \{ \|x_T - x_T^*\|, \|x_T + x_T^*\| \}. \quad (22)$$

The distance is measured in the full activity space, so departures from the teacher memory manifold contribute to it. If other states attain the same maximum, we use the distance to the nearest ideal teacher state, as described in Section 6.5.2.

For each fixed teacher and student-weight configuration, we initialize teacher activity from many random unit states and student activity from zero, then let both populations relax under the full coupled dynamics. We record the teacher trajectory and its final distance from an ideal state. Figure 8A will show representative trajectories with the ideal states marked, together with box-and-whisker plots of final distances across initializations, independent networks, and student learning stages. Small final distances would support Hypothesis 1. Both populations must satisfy the settling criteria; runs that fail to settle are retained and identified. The full protocol is given in Section 6.5.3.

Testing Hypothesis 2: student plasticity approximates descent on the maximum. Hypothesis 2 predicts the reduction in $\mathcal{F}_{\max}$ produced by a small student-plasticity event. After teacher and student activity have relaxed at fixed weights, the observed student state x_S and its recurrent prediction error ε_S determine the local weight change,

$$\Delta W_S = h\pi_S P_0[\varepsilon_S x_S^\top], \quad \varepsilon_S = M_S x_S. \quad (23)$$

Here $h = \eta\Delta t_{\text{pl}}$, as defined above. If this update approximates a gradient-descent step on $\mathcal{F}_{\max}$, its predicted first-order effect is

$$\underbrace{\mathcal{F}_{\max}(W_S; W_T) - \mathcal{F}_{\max}(W_S + \Delta W_S; W_T)}_{\text{observed reduction}} \approx \underbrace{\frac{\|\Delta W_S\|_F^2}{h}}_{\text{predicted reduction}}. \quad (24)$$

This follows because a gradient step reduces a differentiable objective, to first order, by the step size times the squared norm of its admissible gradient. Expressing that gradient through ΔW_S gives the right-hand side of Eq. (24).

The protocol uses the same teacher networks, frozen student-weight configurations, and activity initializations as the test of Hypothesis 1. For each relaxation bout, we calculate $\mathcal{F}_{\max}$ before plasticity and the update from the observed student activity and error. We then apply one discrete update, keeping W_T fixed, and recompute $\mathcal{F}_{\max}$ from the updated student weights using Eq. (56). This recalculation includes the student's new equilibrium response to every candidate teacher state and permits a different teacher state to attain the maximum. The test therefore measures the change in the maximum over the entire teacher memory manifold, rather than only the change in energy at the previously selected teacher state.

We repeat each event with decreasing values of h , always restoring the same pre-event weights and using the same relaxed activity states. Figure 8B will plot the observed reduction against the predicted reduction; agreement corresponds to the diagonal. For comparisons across event sizes, both axis values are divided by h , so smaller updates do not produce apparent agreement merely by moving all points toward zero. The teacher's distance from an ideal state and the settling checks are retained to assess whether discrepancies persist when the activity conditions of the hypothesis are met. Runs that fail to settle remain identified in the reported ensemble.

The same comparison is used for every event, including events that leave the maximum unchanged or increase it. Agreement would support the predicted effect of local gradient descent on $\mathcal{F}_{\max}$; it would not uniquely establish the direction of the full weight update.

Tracking the largest settled student free energy over redistribution cycles. The third panel will test whether repeated teacher-state selection and student plasticity reduce $\mathcal{F}_{\max}$ over successive redistribution cycles. We calculate $\mathcal{F}_{\max}$ before and after each plasticity event, using the updated student weights. The final teacher and student states of one cycle initialize their respective activities in the next.

A matched reference condition will draw a fresh teacher state uniformly from $\mathbb{S}_T$, hold that teacher state fixed while the student relaxes to its equilibrium response, and apply the same local student-plasticity rule with the same effective step size. Both conditions start from identical weights and are compared by event count; relaxation times are recorded separately. This comparison tests whether autonomous teacher-state selection accelerates the reduction of the largest settled student free energy relative to learning from uniformly sampled teacher states.

The algebraic characterization of complete redistribution can also be retained:

$$\mathcal{F}_{\max} = 0 \iff \mathcal{U}_T \subseteq \ker M_S. \quad (25)$$

Indeed, an eigenvalue $\mu \geq 0$ of S_S gives the eigenvalue $\pi_S \mu / (\pi_T S + \pi_S \mu)$ of N_S , so $\ker N_S = \ker M_S$. At this endpoint, the settled response to every teacher-manifold state satisfies $x_S^* = x_T$, $\varepsilon_{TS}^* = 0$, and $\varepsilon_S^* = 0$. The corresponding interface drive and plasticity signal vanish. This identity characterizes the endpoint; reaching it remains an empirical question for the weight dynamics.

The ensemble, settling criteria, parameter sweeps, and numerical accuracy checks for these proposed tests are specified in Materials and methods, Section 6.5.3.

[figure placeholder]

A. Hypothesis 1: teacher-state selection.

Representative teacher trajectories from several initial states, with the independently calculated ideal teacher states marked. Alongside, show box-and-whisker plots of final distances to the nearest ideal state, with individual observations and separate groups for student learning stages or timescale conditions. Identify runs that fail to settle. All distances use the full teacher activity vector.

[figure placeholder]

B. Hypothesis 2: student synaptic plasticity.

Plot the measured reduction in the largest settled student free energy against the prediction $\|\Delta W_S\|_F^2/h$, with a diagonal marking agreement. Repeat at smaller event sizes from identical pre-event weights and activity states. For the step-size comparison, divide both axis values by h and distinguish event sizes by color. Retain events with zero or negative observed reduction.

[figure placeholder]

C. Do repeated cycles reduce the remaining deficits?

Largest settled student free energy versus plasticity-event count, comparing autonomous teacher-state selection with uniformly sampled teacher-manifold states under matched student plasticity. Show variation across independent memory sets and retain plateaus or increases if observed.

Fig 8. Proposed numerical tests of a minimax interpretation of memory redistribution.

This figure is a placeholder; no simulation outcomes are represented. **A**, Hypothesis 1: teacher trajectories and final distances to an ideal teacher state that maximizes the settled student free energy. Both signs of a maximizing direction are equivalent; distances are measured to the nearest ideal state. **B**, Hypothesis 2: the reduction in the largest settled student free energy after one local student-weight update will be compared with $\|\Delta W_S\|_F^2/h$, the prediction for a small gradient-descent step. The maximum will be recomputed after each event, and both reductions expressed per unit step when comparing event sizes. This tests the effect on the objective, rather than the full update direction. **C**, Repeated cycles will track the largest settled student free energy, with matched learning from randomly sampled teacher states as a reference. Reference-state calculations and protocols are given in Materials and methods, Sections 6.5.1–6.5.3.

3.4 Minimizing the largest neocortical reconstruction error

The preceding results show that the coupled dynamics can redistribute memory content from the teacher to the student. We now ask whether this process can be understood as the local optimization of a single quantity at the timescale of consolidation.

As defined in Eq. (8), the student free energy F_S combines the student's recurrent prediction error with the mismatch between the teacher and student states. For fixed teacher activity x_T and fixed student weights W_S , student activity follows gradient descent on F_S . Once the student has settled, the remaining free energy is therefore

$$\min_{x_S} F_S(x_S, x_T; W_S).$$

We analyze the network under the separation of timescales

$$\tau_S \ll \tau_T \ll \frac{1}{\eta}. \quad (26)$$

The first inequality allows the student activity to settle while the teacher activity changes little. The second allows both activities to converge before an appreciable change in the student weights. During this process, teacher activity is restricted to the unit sphere of its memory manifold,

$$\mathbb{S}_T = \{x_T \in \mathcal{U}_T : \|x_T\| = 1\}. \quad (27)$$

We show in this section that, once the student has settled and under the assumptions specified in the formal analysis, the teacher dynamics approach a state x_T^* satisfying

$$x_T^* \in \arg \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_S(x_S, x_T; W_S). \quad (28)$$

The teacher dynamics therefore select a state for which the settled student free energy is maximal.

This nested quantity also has an interpretation in terms of the complete teacher–student network free energy

$$F_{\text{net}}(x_T, x_S; W_T, W_S) = F_T^{\text{rec}}(x_T; W_T) + F_S(x_S, x_T; W_S). \quad (29)$$

Since $\mathcal{U}_T = \ker M_T$, the teacher recurrent free energy vanishes for every $x_T \in \mathbb{S}_T$. Consequently,

$$\max_{x_T \in \mathbb{S}_T} \min_{x_S} F_{\text{net}}(x_T, x_S; W_T, W_S) = \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_S(x_S, x_T; W_S). \quad (30)$$

The largest settled student free energy over the teacher memory manifold is therefore also the largest settled free energy of the complete network over this manifold.

Once the teacher and student activities have converged a student-plasticity event is performed. Under the same timescale separation and the additional assumptions stated below, we show that this event correspond to the gradient-descent step

$$\Delta W_S = -\eta \nabla_{W_S} \left[\max_{x_T \in \mathbb{S}_T} \min_{x_S} F_{\text{net}}(x_T, x_S; W_T, W_S) \right]. \quad (31)$$

Thus, the network dynamics perform greedy minimization over the largest settled free energy of the complete network over the teacher memory manifold. We call redistribution cycle the series of events correspondding to the convergence of the student, the teacher and finally the student synaptic plasticity.

For the formal analysis, we introduce the shorter notation

$$q(x_T; W_S) = \min_{x_S} F_{\text{net}}(x_T, x_S; W_T, W_S) \quad x_T \in \mathbb{S}_T, \quad (32)$$

and

$$\mathcal{F}_{\text{max}}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} q(x_T; W_S). \quad (33)$$

The dependence of q on the fixed teacher weights is omitted from its arguments because the teacher recurrent free energy is zero on $\mathbb{S}_T$.

By definition,

$$q(x_T; W_S) \leq \mathcal{F}_{\max}(W_S; W_T) \quad \text{for every } x_T \in \mathbb{S}_T.$$

Thus, $\mathcal{F}_{\max}$ bounds the settled free energy of the complete network over the entire teacher memory manifold. A low value means that the network settles to a low free energy for every state on this manifold, whereas a large value means that the settled free energy remains high for at least one state. In this notation, Eq. (31) becomes

$$\Delta W_S = -\eta \nabla_{W_S} \mathcal{F}_{\max}(W_S; W_T). \quad (34)$$

This construction is close in purpose to classical predictive coding, while differing in how the states on which free energy is minimized are obtained. PC networks ordinarily minimize free energy for activity elicited by current sensory input. Exploration of the dataset is induced by variation of the such inputs. During autonomous redistribution, no external input specifies the state on which plasticity should act. Therefore the network dynamics has to orchestrate both local free energy minimization and exploration of the relevant data i.e the teacher memory manifold.

The coupled architecture can therefore be interpreted as applying a PC-like objective across a memory manifold rather than to a state selected by current sensory input. The comparison concerns the quantity reduced by the dynamic at the end of a redistribution cycle and does not imply that the instantaneous network dynamics minimize the complete-network free energy as for classical PC.

We now determine what each stage of a redistribution cycle optimizes. The analysis follows the separation of timescales in Eq. (26):

1. For fixed teacher activity and fixed synaptic weights, student relaxation reaches the unique global minimum of the network free energy with respect to x_S .
2. With the student maintained at this equilibrium and the synaptic weights fixed, teacher relaxation reaches the global maximum of the settled network free energy over $\mathbb{S}_T$.
3. Once both activities have converged, student plasticity produces a local decrease in this maximum with respect to W_S .

The first two results concern global optimization over neuronal activity. The third concerns the effect of one sufficiently small synaptic update. It does not imply that the complete weight trajectory reaches a global minimum of $\mathcal{F}_{\max}$.

The separation $\tau_S \ll \tau_T$ permits adiabatic elimination of the student activity. In the limit $\tau_S/\tau_T \rightarrow 0$, the student remains at its equilibrium $x_S^*(x_T)$ while the teacher evolves. The second separation, $\tau_T \ll 1/\eta$, keeps the student weights effectively fixed during teacher selection. The slow plasticity dynamics can therefore be evaluated at the converged teacher and student states. The reduced sequence is

$$x_S \longrightarrow x_S^*(x_T), \quad x_T \longrightarrow x_T^*, \quad W_S \longrightarrow W_S + \Delta W_S. \quad (35)$$

The analysis assumes that teacher activity belongs to

$$\mathbb{S}_T = \{x_T \in \ker M_T : \|x_T\| = 1\}.$$

The two parts of this constraint are implemented differently in the simulations. Renormalization after every integration step enforces $\|x_T\| = 1$ exactly. Membership in $\ker M_T$ is approximate because noise and the interface drive can produce off-manifold activity, while the teacher recurrent dynamics damp this activity.

Let $\sigma_{\min}^2$ denote the smallest nonzero eigenvalue of $M_T^\top M_T$. The teacher recurrent term damps activity outside the memory manifold at a rate of at least $\pi_T \sigma_{\min}^2 / \tau_T$. We choose the effective teacher-side interface gain so that

$$\kappa < \pi_T \sigma_{\min}^2, \quad (36)$$

placing the interface gain below the strength of recurrent restoring forces outside $\mathcal{U}_T$. The MNIST transfer simulations use

$$\tau_S = 1, \quad \tau_T = 10, \quad \eta = 0.005, \quad \kappa = 0.45 \pi_T \sigma_{\min}^2. \quad (37)$$

In a recorded 196-unit MNIST transfer simulation with these parameters, the mean ratio of the norm of the teacher activity projected onto $\mathcal{U}_T$ to its full norm was 0.986. This provides numerical support for approximate confinement in that run.

Report confinement for the current figure run and add the timescale, interface-gain, and noise sweeps to S3 Fig. and Materials and methods before claiming validation across parameter regimes.

We first eliminate the student activity. Define

$$S_S = M_S^\top M_S, \quad N_S = \pi_S S_S (\pi_{TS} I + \pi_S S_S)^{-1}. \quad (38)$$

Proposition 1 (Global student equilibrium). *For fixed $x_T \in \mathbb{S}_T$ and fixed W_S , the linear student dynamics converge from every initial state to the unique global minimizer*

$$x_S^*(x_T) = \arg \min_{x_S} F_{\text{net}}(x_T, x_S; W_T, W_S) = (I - N_S)x_T. \quad (39)$$

The network free energy remaining after student relaxation is

$$q(x_T; W_S) = \frac{\pi_{TS}}{2} x_T^\top N_S x_T. \quad (40)$$

Moreover,

$$q(x_T; W_S) = 0 \iff M_S x_T = 0. \quad (41)$$

The student minimum is global because its Hessian is positive definite:

$$\nabla_{x_S}^2 F_S = \pi_{TS} I + \pi_S M_S^\top M_S \succ 0.$$

The student free energy is therefore strictly convex in x_S , and its dynamics converge to a single equilibrium rather than merely approaching a stationary point.

At this equilibrium, the interface error is

$$\varepsilon_{TS}^* = x_T - x_S^* = N_S x_T. \quad (42)$$

The vector $N_S x_T$ is the part of the current teacher state that the student fails to reproduce after its activity has settled. If $N_S x_T = 0$, the student reproduces the teacher state exactly and the settled network free energy vanishes. If it is nonzero, its magnitude and direction describe the discrepancy that remains available to influence teacher selection.

We next determine which teacher state maximizes this remaining free energy. Let $U_T \in \mathbb{R}^{d \times r}$ have orthonormal columns spanning the nonzero teacher memory space $\mathcal{U}_T$, and define

$$A_S = U_T^\top N_S U_T. \quad (43)$$

The operator A_S compares the unresolved student errors along the directions that the teacher can express. Its leading eigenspace identifies the teacher-supported direction with the largest remaining student deficit.

Proposition 2 (Global teacher selection). *For fixed W_S , the largest settled network free energy is*

$$\mathcal{F}_{\max}(W_S; W_T) = \frac{\pi_{TS}}{2} \lambda_{\max}(A_S). \quad (44)$$

The maximizing teacher states are the unit states whose coordinates in the basis U_T belong to the leading eigenspace of A_S .

Consider the projected teacher dynamics

$$\tau_T \dot{x}_T = \kappa P_{T,x} \varepsilon_{TS}^*, \quad P_{T,x} = U_T U_T^\top - x_T x_T^\top, \quad (45)$$

where $P_{T,x}$ projects activity changes onto the tangent space of $\mathbb{S}_T$. For $\kappa > 0$, if the initial teacher state has a nonzero component in the leading eigenspace of A_S , these dynamics converge to a global maximizer of q over $\mathbb{S}_T$.

If the largest eigenvalue is simple, with unit eigenvector v_1 , the two maximizing states are

$$x_T^* = \pm U_T v_1. \quad (46)$$

The form and sign of the teacher-side interface coupling follow from this optimization. Because x_S^* minimizes the network free energy with respect to x_S , changes in x_S^* do not add a term to the derivative of the settled value. Along allowed teacher-state changes, the gradient is therefore

$$P_{T,x} \nabla_{x_T} F_{\text{net}}|_{x_S=x_S^*} = \pi_{TS} P_{T,x} \varepsilon_{TS}^*. \quad (47)$$

The projected teacher dynamics consequently perform gradient ascent on q :

$$\frac{dq}{dt} = \frac{\kappa \pi_{TS}}{\tau_T} \|P_{T,x} \varepsilon_{TS}^*\|^2 \geq 0. \quad (48)$$

Thus, the settled network free energy cannot decrease during teacher relaxation. Under the initialization condition in Proposition 2, teacher activity converges to the direction for which this settled value is globally maximal.

This result also explains the effective top-down inhibition in the model. The interface error is

$$\varepsilon_{TS}^* = x_T - x_S^*.$$

Student activity therefore subtracts from the drive received by the teacher. A teacher component already reproduced by the student is cancelled from the interface error, whereas an unmatched component continues to drive teacher activity. The teacher dynamics consequently move toward content that remains poorly represented by the student.

After the student has reached its global minimum and the teacher has reached its global maximum, we evaluate the student plasticity rule at the selected state.

Theorem 2 (Local reduction of the largest settled network free energy). *Assume that the largest eigenvalue of A_S is simple and that both activities have converged to*

$$x_T = x_T^*, \quad x_S = x_S^*(x_T^*).$$

The gradient of the largest settled network free energy with respect to the student weights is then

$$\nabla_{W_S} \mathcal{F}_{\text{max}} = -\pi_S \varepsilon_S^* x_S^{*\top}, \quad \varepsilon_S^* = M_S x_S^*. \quad (49)$$

For unconstrained student weights, one plasticity event gives

$$\Delta W_S = \eta \pi_S \varepsilon_S^* x_S^{*\top} = -\eta \nabla_{W_S} \mathcal{F}_{\text{max}}. \quad (50)$$

For sufficiently small η ,

$$\mathcal{F}_{\text{max}}(W_S + \Delta W_S; W_T) = \mathcal{F}_{\text{max}}(W_S; W_T) - \eta \|\nabla_{W_S} \mathcal{F}_{\text{max}}\|_F^2 + o(\eta). \quad (51)$$

The plasticity direction therefore gives the greatest first-order decrease in the current value of $\mathcal{F}_{\text{max}}$ among weight changes of the same Frobenius norm.

The simple-eigenvalue assumption ensures that the maximizing teacher direction is locally unique up to sign. Reversing the teacher state also reverses the settled student activity and its recurrent error:

$$x_T^* \longrightarrow -x_T^*, \quad x_S^* \longrightarrow -x_S^*, \quad \varepsilon_S^* \longrightarrow -\varepsilon_S^*.$$

Their outer product is unchanged,

$$(-\varepsilon_S^*)(-x_S^*)^\top = \varepsilon_S^* x_S^{*\top},$$

so the two maximizing teacher states produce the same weight gradient.

When self-connections are excluded, let P_0 denote the operation that sets the diagonal entries of a matrix to zero. The implemented student update is then

$$\Delta W_S = -\eta P_0 \nabla_{W_S} \mathcal{F}_{\max}. \quad (52)$$

Its first-order effect is

$$\mathcal{F}_{\max}(W_S + \Delta W_S; W_T) = \mathcal{F}_{\max}(W_S; W_T) - \eta \|P_0 \nabla_{W_S} \mathcal{F}_{\max}\|_F^2 + o(\eta). \quad (53)$$

The projected update therefore gives the steepest first-order decrease among admissible weight changes that preserve the absence of self-connections.

These results establish the optimization performed during one redistribution cycle. Student relaxation globally minimizes the network free energy over x_S . Teacher relaxation then globally maximizes the settled value over $\mathbb{S}_T$. Finally, student plasticity locally decreases this maximum with respect to W_S :

$$\min_{x_S} \longrightarrow \max_{x_T \in \mathbb{S}_T} \longrightarrow -\nabla_{W_S} \mathcal{F}_{\max},$$

with the weight gradient projected onto the admissible synapses when self-connections are excluded. Because plasticity changes W_S , it also changes N_S and A_S . The direction that maximizes the student deficit during the next redistribution cycle may therefore differ from the direction selected during the current cycle.

Complete redistribution is characterized by

$$\mathcal{F}_{\max} = 0 \iff \mathcal{U}_T \subseteq \ker M_S. \quad (54)$$

At this endpoint,

$$x_S^* = x_T, \quad \varepsilon_{TS}^* = 0, \quad \varepsilon_S^* = 0$$

for every $x_T \in \mathbb{S}_T$. The signals driving both teacher selection and student plasticity therefore vanish.

These results describe one adiabatically separated redistribution cycle. They do not prove that the complete weight trajectory reaches $\mathcal{F}_{\max} = 0$ from arbitrary initial conditions. Complete derivations, including the treatment of repeated leading eigenvalues and the projection onto zero-diagonal weights, are provided in Materials and methods. The same section describes the finite-timescale tests, teacher-manifold occupancy measure, and parameter sweeps used to assess the adiabatic-elimination and hard-manifold approximations.

4 Discussion

However, CA1-PFC representational coordination was weaker and less faithful to the waking experience during sleep than during awake ripples. The authors interpreted sleep reactivation as more integrative and less like precise reinstatement. CLEAR LIMITATION DUE TO THE FACT THAT WE USES SIMPLE ASSOCIATIVE MEMORY NETWORKS

4.1 Summary

[One paragraph: recipient deficit selects teacher content; selected content trains the recipient; learning removes the selecting signal. Prioritization, reduced reactivation of consolidated content, and self-extinction follow from the same objective.]

4.2 Relationship to other models

[Compare utility-based replay, context-driven replay, autonomous attractor consolidation, generative consolidation, recall-gated consolidation, and Go-CLS by their priority variable, autonomous selection mechanism, and what is transferred. The narrow novelty claim is a closed loop from recipient-deficit objective to source selection and recurrent recipient storage.]

4.3 Awake prefrontal memory control may be reused during sleep

[Lead with awake retrieval suppression and retrieval-induced forgetting. During wake, PFC control can suppress unwanted or competing hippocampal retrieval. Propose that related effective circuitry is reused offline: behavioral goals select during wake, whereas recipient competence selects during consolidation. Link to NREM PFC-associated suppression of CA1 activity and reactivation as evidence for a suppressive motif, not proof that PFC computes $\mathcal{D}_{\max}$.]

4.4 Experimental predictions, limitations, and future directions

[Predictions: cortical similarity suppresses matching hippocampal reactivation after controls for source strength, recency, salience, and reward; disrupting top-down suppression increases redundant reactivation and reduces efficiency; priority returns after cortical degradation. Limitations: linear subspaces; restricted ReLU extension; no ordered replay; idealized constraints and timescales; degeneracy; transformed codes; omitted source reliability, context, reward, emotion, salience, and neuromodulation. State explicitly that the analytic efficiency claim is local, that the exact $2/(1-\chi)$ factor assumes unconstrained weights and an already stored common component, and that faster complete transfer is supported numerically rather than by a global convergence or hitting-time theorem. Continual learning is a separate downstream project.]

5 Conclusion

[Two or three sentences. Close on the biological and mathematical message: cortical dependence can organize hippocampal reactivation; effective cortical suppression of the teacher follows from maximizing recipient deficit; and the discrepancy-driven prioritization signal vanishes at complete transfer.]

6 Materials and methods

6.1 Memory construction and network initialization

[Pattern generation, covariance-memory construction, dimensions, rank, correlation, recipient familiarity, seeds, and no-autapse constraint.]

6.2 State dynamics and diagnostics

[Teacher and recipient equations, activation, integration scheme, timestep, timescales, coupling, noise, normalization, settling, stopping, manifold leakage, and terminal occupancy.]

6.3 Prioritized, oracle, and random protocols

[Executable pseudocode or an algorithm box. Matching initializations, candidate states, one plasticity event per bout, thresholds, seed count, and uncertainty summaries.]

6.4 Outcome measures

[Define N_S , A_S , $\mathcal{D}_{\max}$, errors, occupancy, cumulative weight change, threshold events, oracle gap, nearest-memory distance, and cone confinement.]

6.5 Reference states and numerical tests of redistribution

The calculations below define the ideal teacher states and free-energy values used in Figure 8. The accompanying simulation protocol is proposed; the measurements have not yet been performed.

6.5.1 Calculating an ideal teacher state

The ideal teacher state depends on the teacher memory space and the current student weights. We calculate it while both weight matrices are fixed. Let $U_T \in \mathbb{R}^{d \times r}$ have orthonormal columns spanning $\mathcal{U}_T = \ker M_T$. These columns provide coordinates for the teacher memory space: every allowed teacher state can be written as $x_T = U_T a$, with $\|a\| = 1$. The basis will be obtained from the right singular vectors of M_T associated with zero singular values. The numerical rank tolerance will be stated, and the basis checked for orthonormality and for $M_T U_T$ being negligible at that tolerance.

The novelty matrix N_S , defined in Eq. (17), determines the student's settled free energy for any teacher state. To compare only states within the teacher memory space, we form

$$A_S = U_T^\top N_S U_T. \quad (55)$$

This smaller matrix describes how the student's remaining error varies among combinations of teacher memories. Equation (19) gives

$$q(U_T a; W_S) = \frac{\pi T S}{2} a^\top A_S a, \quad \mathcal{F}_{\max} = \frac{\pi T S}{2} \lambda_{\max}(A_S). \quad (56)$$

To see why the largest eigenvalue gives the maximum, expand the unit vector a in an orthonormal eigenbasis of the symmetric matrix A_S . Its quadratic form is then a weighted average of the eigenvalues, with nonnegative weights summing to one. The largest possible value is the largest eigenvalue, attained by any unit vector in its eigenspace. Thus, for a leading unit eigenvector v_1 , an ideal teacher state is

$$x_T^* = U_T v_1. \quad (57)$$

Its negative is equally valid. The corresponding student state is $x_S^*(x_T^*) = (I - N_S)x_T^*$. This calculation uses only the fixed weights and precisions; the simulated trajectory is not used to choose the reference state, and the reference state does not drive the simulation.

After each student-plasticity event, N_S and A_S are recalculated from the updated W_S , while U_T remains fixed. Matrix inverses in the analytical expressions will be evaluated through linear solves.

6.5.2 Distance to an ideal teacher state

When the leading eigenvalue of A_S is simple, the ideal states are x_T^* and $-x_T^*$, and Eq. (22) gives the distance to the closer state. The observed teacher activity is already normalized by the network dynamics. We use this full activity vector for the distance calculation, without projecting it onto the memory manifold. The distance therefore reflects both departure from the memory manifold and a difference in direction within it.

If several independent directions attain the maximum, no single reference direction is preferred. Let V_1 contain an orthonormal basis of the leading eigenspace of A_S . We then use the distance to the nearest ideal state,

$$d_T = \min_{\|b\|=1} \|x_T - U_T V_1 b\| = \sqrt{2 - 2\|V_1^\top U_T^\top x_T\|}. \quad (58)$$

The equality uses $\|x_T\| = 1$ and orthonormality of $U_T V_1$. It follows by choosing the unit reference with the largest inner product with x_T . For a one-dimensional leading eigenspace, this reduces to choosing

the closer sign in Eq. (22). Numerical equality of eigenvalues will be determined using a stated precision tolerance; distinct eigenvalues resolved at that precision will remain distinct.

State distance is a stricter test than proximity to the maximum free energy. When the two largest eigenvalues are close, teacher states far from an exact maximizer may still yield almost the same settled student free energy. The leading eigenvalue gap will therefore be recorded when interpreting final distances. For the empty student, $W_S = 0$, every unit state on the teacher memory manifold is ideal. This condition is reported separately as a control for confinement, since it cannot demonstrate selection of a particular memory direction. Partially trained students provide the unequal costs needed to test directional selection.

6.5.3 Proposed simulation and reporting protocol

The baseline ensemble will use the linear model with zero-diagonal recurrent weights, $d = 48$ units per population and teacher-memory rank $r = 6$. The proposed starting design is 20 independently generated teacher memory sets, 10 activity initializations per frozen student configuration, and four weight snapshots spanning partial redistribution from separate calibration runs. Snapshot times are fixed before the selection tests, and the empty-student control is reported separately. Independent memory sets form the replication units; repeated initializations and weight snapshots within a set are treated as grouped observations. These sample counts are planning choices, not completed experiments.

For each fixed pair of weight matrices, the ideal teacher states will first be calculated as above. Initial teacher states will be drawn uniformly from the full unit activity sphere by normalizing independent isotropic Gaussian vectors. Student activity will start at zero. Both populations will then evolve under Eqs. (6) and (7), with plasticity disabled during relaxation. The teacher retains its recurrent restoring force and unit-norm normalization; no projection onto the teacher memory manifold is applied.

The noiseless run will stop when the teacher’s normalized deterministic drift and the student’s departure from its equilibrium response both remain below prescribed tolerances throughout an observation interval. The teacher drift used here is the component of its deterministic activity derivative tangent to the unit sphere, because normalization removes the radial component. Student settling is checked by comparing the observed x_S with $x_S^*(x_T)$ for the current teacher state. These criteria are independent of distance to an ideal state: a trajectory can settle and still select the wrong direction. A maximum run duration is imposed, and final distances from runs that fail to settle are retained with a failure indicator.

For each condition, panel A will summarize the final distances with boxes showing the median and interquartile range, whiskers extending to the most extreme observations within 1.5 interquartile ranges, and individual observations shown so that failures and extreme distances remain visible. Student learning stages and timescale conditions will be displayed separately. Uncertainty estimates will group observations by independent teacher memory set. Representative trajectories will use a teacher memory space of rank two. Only the display uses memory-space coordinates; the projected trajectories are not renormalized, and all reported distances are calculated in the full activity space.

The baseline coefficients will be $\pi_T = 1$, $\pi_S = 0.5$, $\pi_{TS} = 1$, $\tau_S = 1$, and $\tau_T = 10$. Teacher-side gain will be scaled as $\kappa = 0.45\pi_T\sigma_{\min}^2$, where $\sigma_{\min}^2$ is the smallest nonzero eigenvalue of $M_T^\top M_T$. This places the gain below the recurrent restoring scale outside the memory manifold without imposing exact confinement. The noiseless event protocol will be followed by sweeps of τ_S/τ_T , $\kappa/(\pi_T\sigma_{\min}^2)$, memory rank and correlation, and plasticity step size. Numerical tolerances, snapshot times, integration step, maximum relaxation duration, and sweep grids will be fixed after a pilot and documented with the final figure. A smaller integration step will be used to check that the measured discrepancies reflect the dynamics rather than discretization.

For the single-event test of Hypothesis 2, each relaxed configuration is copied before applying updates of different sizes. The local update uses the observed student activity and recurrent error. The reference maximum is recomputed after each update using Eq. (56). The first-order prediction in Eq. (24) assumes that $\mathcal{F}_{\max}$ is locally differentiable. The finite changes remain measurable without this assumption, so all configurations are retained in the same comparison. Event sizes must be small enough to test the first-order prediction while producing changes larger than the numerical uncertainty in $\mathcal{F}_{\max}$. Events with negligible updates are identified separately, since agreement at zero does not test the plasticity hypothesis.

Persistent-noise conditions will measure the distribution of distances to the nearest ideal teacher state over a fixed observation window rather than claim convergence to a fixed state. Separate runs with simultaneous activity and plasticity will assess whether the cycle-based interpretation remains useful for the continuous protocol used elsewhere in the paper. Results will include failures and variation across memory sets. No convergence or gradient-descent claim for these simulations is made before those measurements are available.

6.6 Structured-memory transfer comparison

6.6.1 Controlled imbalanced corpus

The fast-transfer experiments used $P = 26$ unit-normalized memories arranged into three mutually orthogonal groups of sizes $n_g \in \{18, 5, 3\}$. Each group occupied a three-dimensional subspace, so the complete teacher memory space had rank $r = 9$. We assigned eight neurons to each independent direction, giving $d = 72$ teacher and student units. A fixed permutation of neuron coordinates removed any visible block structure from the recurrent weights without changing the memory geometry.

Within group g , memory p was constructed as

$$m_{g,p} = B_g c_{g,p}, \quad c_{g,p} = \frac{c_0 + s_g (\cos \theta_{g,p} u + \sin \theta_{g,p} v)}{\sqrt{1 + s_g^2}}, \quad \theta_{g,p} = \frac{2\pi p}{n_g} + \phi_g, \quad (59)$$

where the columns of B_g form an orthonormal basis for group g , and c_0, u, v are orthonormal vectors in its three-dimensional coefficient space. The group phases ϕ_g were fixed across conditions. Evenly spaced residual directions satisfy a mean off-diagonal inner product of $-1/(n_g - 1)$, so choosing

$$s_g^2 = \frac{1 - \rho}{\rho + 1/(n_g - 1)} \quad (60)$$

sets the mean cosine similarity between distinct memories in every group to ρ . The main comparison used $\rho = 0.70$. The correlation sweep used $\rho \in \{0, 0.35, 0.70\}$ while keeping the group bases, sizes, phases, rank, network parameters, and random seeds fixed. The $\rho = 0$ construction contains signed memories; all correlation-sweep simulations therefore used linear units.

The Gram matrix in Fig. 6(A) is $G = \mathcal{M}^\top \mathcal{M}$, where $\mathcal{M}$ contains the 26 memories as columns. Because every memory has unit norm, G_{pq} is their cosine similarity. The three diagonal blocks show within-group redundancy, whereas the zero-valued off-block entries follow from orthogonality between groups.

6.6.2 Matched transfer protocols

We constructed the frozen teacher weights from the same 26-memory corpus in both conditions. Both students started with $W_S = 0$ and used linear units, plasticity rate $\eta = 0.01$, integration interval $\Delta t = 0.05$, and one projected recurrent-weight update per plasticity event. The diagonal of W_S was reset to zero after every update. Each run contained 200,000 events. The state time constants were $\tau_S = 0.1$ and $\tau_T = 1$, the gains were $\pi_T = 1$, $\pi_S = 0.5$, $\pi_{TS} = 1$, and $\pi_{ST} = -0.5$ in the implementation sign convention. Teacher noise had amplitude 0.05, and teacher activity was normalized around unit norm.

In the random-presentation condition, an index was drawn uniformly from the 26 named memories at each event. The selected memory was clamped directly onto the student before applying its recurrent update. In the coupled condition, neither network was clamped. The teacher and student states advanced by one integration step under the coupled dynamics, after which the student received the same local recurrent update. Teacher weights remained fixed in both conditions. Model construction used seed 12, the random presentation schedule used seed 23, and stochastic state evolution used seed 12,357.

Uniform sampling over named memories weights the expected random-presentation update by the empirical second moment

$$C_m = \frac{1}{P} \sum_{p=1}^P m_p m_p^\top. \quad (61)$$

The 18-memory group therefore contributes more repeated updates than the 5- and 3-memory groups, although all three groups add the same number of independent directions. Stronger within-group correlation concentrates those repeated updates further onto the shared component of the large group.

6.6.3 Redistribution measures

For each named memory, we measured the recurrent reconstruction residual

$$r_p(t) = \|m_p - W_S(t)f(m_p)\|_2. \quad (62)$$

The heatmaps in Fig. 6(C) show every $r_p(t)$ in group order. Under the linear activation used here, $f(m_p) = m_p$.

[Link with figure 6 B. and figure 7 :] Let $U_T = [u_1, \dots, u_r] \in \mathbb{R}^{d \times r}$ be an orthonormal basis of the teacher memory space, with $r = 9$, and let $M_S(t) = I - W_S(t)$. We defined the subspace residual as

$$R_{\text{subspace}}(t) = \frac{\|M_S(t)U_T\|_F}{\sqrt{r}} = \left(\frac{1}{r} \sum_{j=1}^r \|M_S(t)u_j\|_2^2 \right)^{1/2}. \quad (63)$$

reconstruct or represent ?

This basis-independent measure gives equal weight to every independent memory space direction. It equals one when $W_S = 0$ and vanishes exactly when the student reconstructs the complete teacher memory space.

We also measured the largest residual over any unit-norm direction in the teacher space,

$$R_{\text{worst}}(t) = \max_{\|a\|_2=1} \|M_S(t)U_T a\|_2 = \sqrt{\lambda_{\max}(U_T^\top M_S(t)^\top M_S(t)U_T)}. \quad (64)$$

This is the largest remaining hole in the student representation, including teacher-supported combinations that do not coincide with a named memory. Figure 6(B) reports both R_{subspace} and R_{worst} . Figure 7 reports only R_{subspace} so that the effect of correlation can be compared on matched axes.

Supporting information

- **S1 Appendix.** Full proofs of Proposition ??, Theorem ??, Corollary ??, and Corollary ??; projected power iteration; envelope steps; antipodal maximizers; feasible projected descent; and the local correlated-memory rate comparison.
- **S1 Fig.** Ascent, zero-drive, and descent teacher-side coupling ablation.
- **S2 Fig.** Rank, dimension, correlation, and recipient-familiarity sweeps.
- **S3 Fig.** Timescale separation, noise, seed dependence, and teacher-manifold leakage.
- **S4 Fig.** Oracle implementation checks and random-baseline controls.
- **S5 Fig.** Rectified-network robustness and time-resolved individual-memory selectivity.

References

- [1] Raichle ME, Gusnard DA. Appraising the brain's energy budget. *Proceedings of the National Academy of Sciences*. 2002;99(16):10237–10239. doi:10.1073/pnas.172399499.
- [2] Raichle ME. The brain's dark energy. *Science*. 2006;314(5803):1249–1250. doi:10.1126/science.1134405.

- [3] Klinzing JG, Niethard N, Born J. Mechanisms of systems memory consolidation during sleep. *Nature Neuroscience*. 2019;22(10):1598–1610. doi:10.1038/s41593-019-0467-3. 752
753
- [4] Staresina BP, Niediek J, Borger V, Surges R, Mormann F. How coupled slow oscillations, spindles and ripples coordinate neuronal processing and communication during human sleep. *Nature Neuroscience*. 2023;26(8):1429–1437. doi:10.1038/s41593-023-01381-w. 754
755
756
- [5] Diekelmann S, Born J. The memory function of sleep. *Nature Reviews Neuroscience*. 2010;11(2):114–126. doi:10.1038/nrn2762. 757
758
- [6] Wang JX, Poe G, Zochowski M. From network heterogeneities to familiarity detection and hippocampal memory management. *Physical Review E*. 2008;78(4):041905. doi:10.1103/PhysRevE.78.041905. 759
760
- [7] Tang M, Barron H, Bogacz R. Recurrent predictive coding models for associative memory employing covariance learning. *PLOS Computational Biology*. 2023;19(4):e1010719. doi:10.1371/journal.pcbi.1010719. 761
762
763